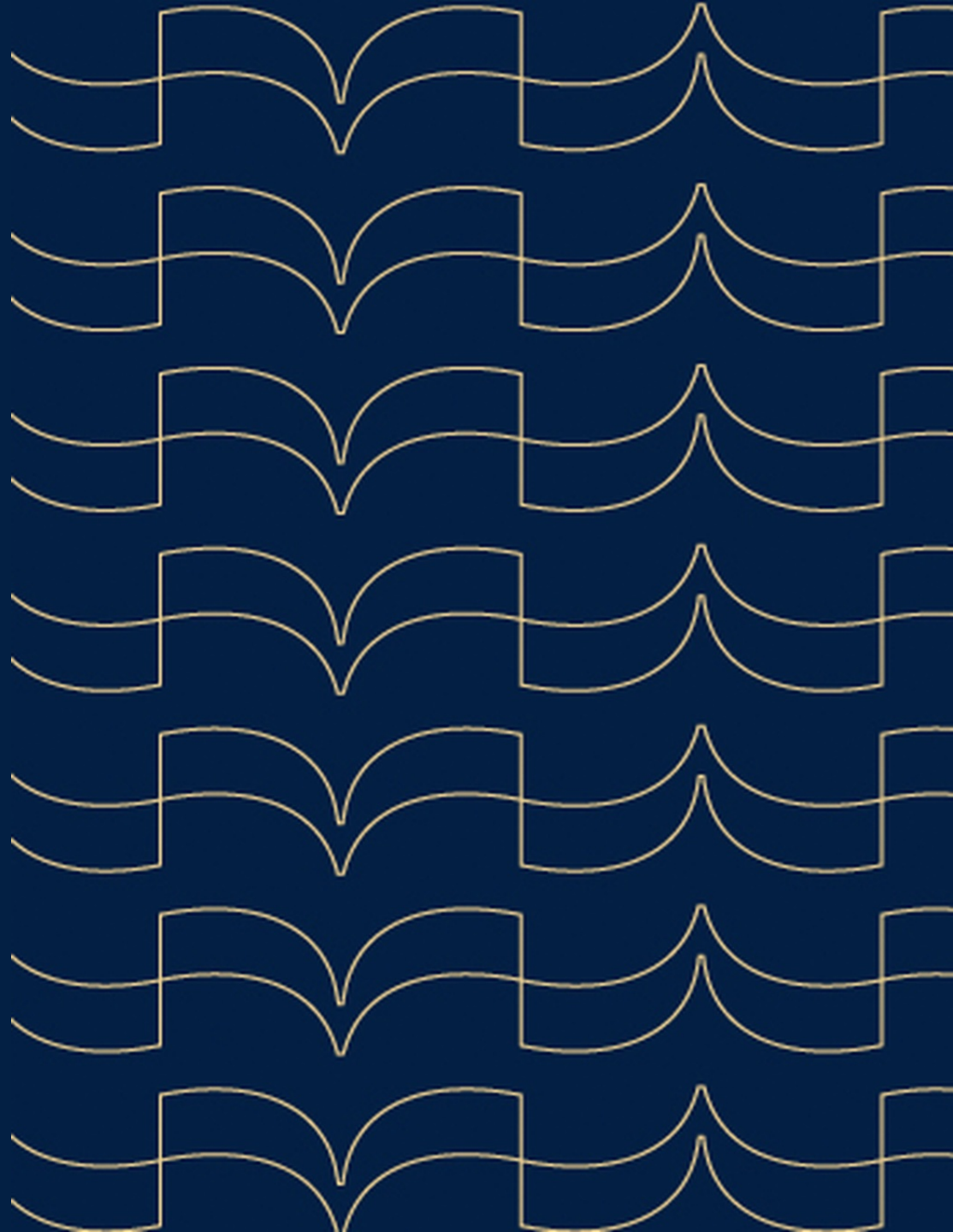
UT SAN ANTONIO

A modernized Space for Hyper Compute

Proposal: **Arc Space** for VS Code

Jason Kolodziej

September 30, 2026



AGENDA

What we'll cover

01

Today

How researchers reach Arc now, per the user guide

02

Friction

Where time, compute and data get lost

03

Proposal

What the extension changes, task by task

04

Next steps

Pilot plan and asks for RCSG

01

How we work on Arc today

TODAY

One session spans a terminal, an editor and the docs

- Log in with SSH and DUO (Terminal, or MobaXterm on Windows), or use the OnDemand web portal.
- Request a compute node by hand: srun -p <partition> -n 1 -t hh:mm:ss -c 40 --pty bash
- Track, inspect and cancel jobs with squeue, scontrol, scancel and sacct in that same shell.
- Move data between /home, /work and /vault with rsync or scp, keeping each tier’s rules in mind.

KEY POINT

Login nodes are for editing, transfers and job submission only. Real work must run on a compute node, so the tools should make that the easy path.

RULES TO REMEMBER

Three policies every Arc user carries in their head

30

Days until /work purge

Files in /work not accessed in 30 days become deletion candidates.

1

Hour of idle GPU allowed

Jobs not using their allocated GPUs are terminated after a 1-hour grace period.

10/20

Jobs / nodes per user

At most 10 active jobs, running on at most 20 compute nodes.

WHY IT MATTERS

The hard part isn’t *Slurm*. It’s juggling tools and rules.



Every switch is a chance to run on a login node, leave a GPU idle, or lose results to the purge.

02

Proposal

THE PROPOSAL

Same cluster, same Slurm, one window

TODAY

Terminal, editor and docs

- Separate windows for shell, editor and portal
- Node requests typed by hand; the session ends with the connection
- Job status, limits and efficiency checked manually

WITH ARC HPC

One VS Code window

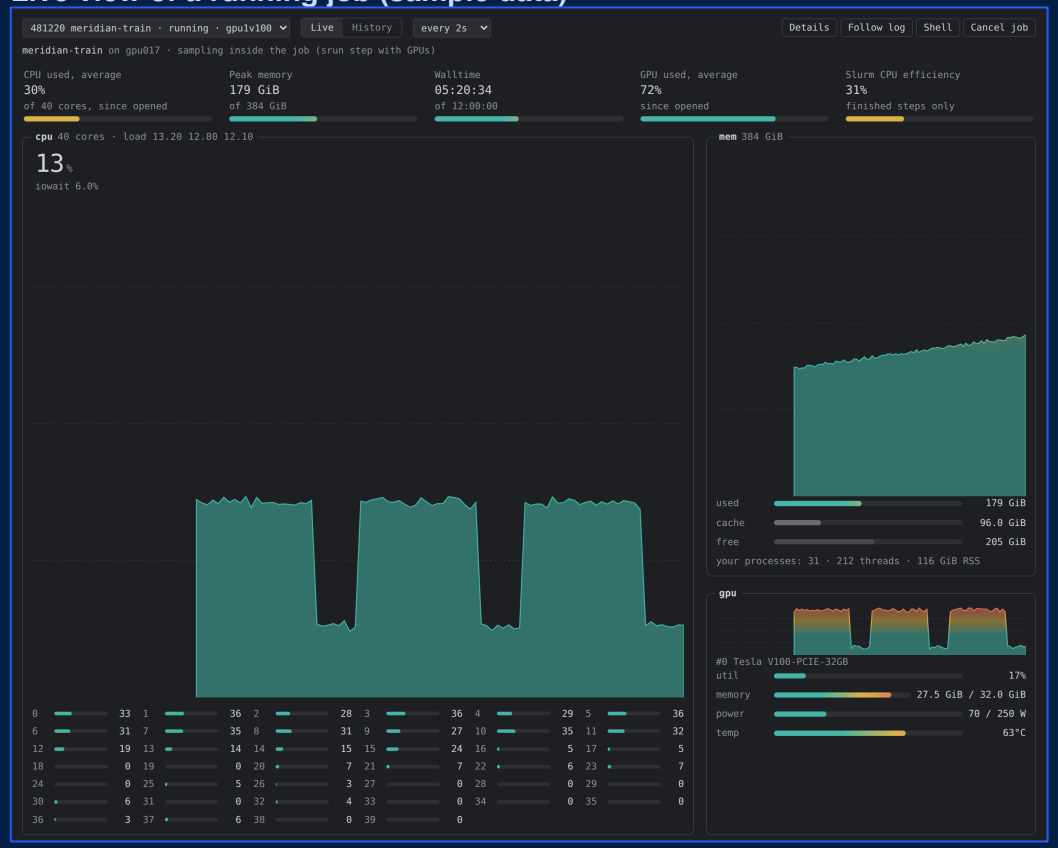
- Start Session: sign in, get a node, reopen this window on it
- Jobs, partitions and storage in one sidebar
- Built-in warnings for walltime, idle GPUs and the /work purge

TASK BY TASK

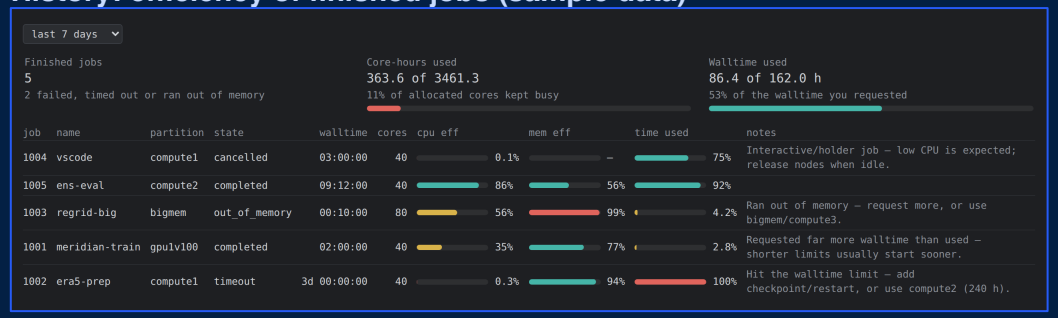
What changes for the researcher

Task	Today (per the user guide)	With Arc HPC
Get onto a compute node	ssh + DUO, then srun … --pty bash	Start Session reopens this window on the node
See which nodes are free	sinfo in a terminal	Partitions view: idle / total, per-node state
Manage running jobs	squeue, scontrol, scancel	My Jobs: details, log, shell, cancel, hold, requeue
Check job efficiency	sacct after the job ends	Live monitor plus history with right-sizing notes
Move data between tiers	rsync or scp by hand	Archive to vault and stage to /work from Explorer
Stay within the rules	Remember them	Purge scan, walltime and idle-GPU warnings, job limits shown

Live view of a running job (sample data)



History: efficiency of finished jobs (sample data)



SPOTLIGHT

See what a job is actually using

A btop-style monitor streams CPU, memory, GPU, network and Lustre I/O from the job's own node, then grades finished jobs on CPU, memory and walltime efficiency from sacct. Sampling runs inside the job's allocation, so it adds nothing to the login nodes.

BUILT FOR ARC

It works within Arc's rules and infrastructure

- Uses only standard SSH, DUO and Slurm; no new services, daemons or privileges on Arc.
- One DUO sign-in per session, reused through an SSH ControlMaster connection.
- Editor and language tooling run on the allocated compute node, not on shared login nodes.
- Vault actions are routed through the login node, where /vault is mounted.

STATUS

Working prototype (v0.4.2), verified on Arc on Sept 30, 2026 by one user: sign-in, node allocation and VS Code on a compute node.

NEXT STEPS

Pilot plan and asks for RCSG

Milestone	Owner	Due	Status
Working prototype (v0.4.2) on Arc	Jason K.	Sep 30	Done
Design and security review	Jason K. + RCSG	TBD	Not started
Confirm node SSH, vault and GPU fit	RCSG	TBD	Not started
Validate job monitor on CPU and GPU nodes	Jason K.	TBD	Not started
Pilot with a small group of Arc users	Jason K.	TBD	Not started

Thank you

Jason Kolodziej

Questions, feedback and pilot volunteers welcome

